

CONTACT INFORMATION	Mailman School of Public Health Columbia University 722 West 168th Street New York, NY 10032	Email: tk3077@cumc.columbia.edu Website: https://taehyeonkoo.github.io/
PROFESSIONAL EXPERIENCE	<i>Postdoctoral Researcher</i> Columbia University Mailman School of Public Health Mentors: Caleb Miles, Kara Rudolph, and Elizabeth Stuart	Jun 2025 - Present
EDUCATION	<i>Doctor of Philosophy</i> , Statistics Rutgers University Advisors: Zijian Guo and Nicole Pashley	May 2025
	<i>Master of Science</i> , Statistics Seoul National University, South Korea Advisor: Johan Lim	Feb 2020
	<i>Bachelor of Science</i> , Mathematical Science Seoul National University, South Korea	Feb 2018
RESEARCH INTERESTS	Causal inference; Partial identification and sensitivity analysis; Design and analysis of randomized experiments; Non-standard inference; Distributionally robust optimization.	
PREPRINTS	Koo, T. , Stuart, E. A., Rudolph, K. E., & Miles, C. H. (2026). Causal Effects of Modified Treatment Policy under Positivity Violations: A Partial Identification Approach. <i>arXiv preprint arXiv:2608.23971</i> .	
	Koo, T. , & Guo, Z. (2026). Synthetic Control with Weight Uncertainty: Robust Identification and Statistical Inference. <i>arXiv preprint arXiv:2511.02632</i> .	
PUBLICATIONS	Koo, T. , & Pashley, N. E. (2026). Design-based Causal Inference for Incomplete Block Designs. <i>Biometrika</i> , asag013.	
	Koo, T. , Lee, Y., Small, D. S., & Guo, Z. (2023). RobustIV and controlfunctionIV: Causal Inference for Linear and Nonlinear Models with Invalid Instrumental Variables. <i>Observational Studies</i> , 9(4), 97-120.	
MANUSCRIPT IN PREPARATION	Koo, T. , Kang, H, Rudolph, K. E., & Miles, C. H. (2026+). Bounding Stratum-Specific Causal Effects under Noncompliance via the Correlation of Potential Treatments, with Application to a Comparative-Effectiveness Trial for Opioid Use Disorder.	
	Koo, T. , & Pashley, N. E. (2026+). Incomplete Block Designs for Multi-Arm Experiments: A Design-Based Guide for Practitioners.	
HONORS AND AWARDS	Early Career Travel Award Joint Statistical Meeting (JSM) 2025	Aug 2025

	Student Travel Award ASA/IMS Spring Research Conference (SRC) 2025	Jun 2025
	Best Ph.D. Qualifying Exam Performance Department of Statistics, Rutgers University	Sep 2021
SOFTWARE	<i>R Packages</i> IBDInfer : A package for design-based inference for incomplete block designs. https://CRAN.R-project.org/package=IBDInfer RobustIV : A package for inference with a possibly invalid instrumental variable in the linear model. https://CRAN.R-project.org/package=RobustIV controlfunctionIV : A package for inference using the control function method in the nonlinear model. https://CRAN.R-project.org/package=controlfunctionIV	
TEACHING EXPERIENCE	<i>Guest Lecturer at Columbia University</i> BIST P9109: Theory of Statistical Inference I	Fall 2026
	<i>Instructor at Rutgers University</i> Review of STAT 593 and 594 for Ph.D. Qualifying Exam	Summer 2022
	<i>Teaching Assistant at Rutgers University</i> STAT 486: Applied Statistical Learning STAT 490: Introduction to Experimental Design STAT 467: Applied Multivariate Analysis STAT 594: Advanced Modern Statistical Inference II STAT 593: Theory of Statistics	Fall 2024 Spring 2024 Spring 2023 Spring 2022 Fall 2021
	<i>Teaching Assistant at Seoul National University</i> 326.311: Mathematical Statistics I 033.019: Introduction to Statistics	Summer 2019 Fall 2018
TALKS AND PRESENTATIONS	<i>Causal Effects of Modified Treatment Policy under Positivity Violations</i> Poster presentation, ACIC 2026 Causal Inference Learning Group Seminar, Columbia University	May 2026 Dec 2025
	<i>Synthetic Control with Weight Uncertainty</i> Invited talk, ASA SLDS Conference 2026 (Upcoming) Invited talk, Korean International Statistical Society (KISS) Webinar Lightning talk, IMS NRC 2026 Poster presentation, ACIC 2026 Contributed talk, JSM 2025 Center for Causal Inference Seminar, University of Pennsylvania	Nov 2026 Sep 2026 Jul 2026 May 2026 Aug 2025 Nov 2024
	<i>Design-based Causal Inference for Incomplete Block Designs</i> Invited talk, ASA/IMS Spring Research Conference 2025 Poster presentation, IMS ICSDS 2024 Poster presentation, Conference on Recent Advances in Statistics and Data Science, Rutgers University	Jun 2025 Dec 2024 May 2023
ACADEMIC SERVICE	<i>Reviewer</i> <i>Journal of Educational and Behavioral Statistics</i>	

REFERENCES

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